La Scuola d'Italia / Dr. Huson / IB Math AA SL
5 May 2026

First & Last name:

11.16 Classwork: Distributions

Problems 1 and 2 are no calculator. Problems 3, 4, 5 calculator allowed.

1. The binomial expansion of $(1+kx)^n$ is given by

$$1+\frac{9x}{2}+15k^2x^2+\cdots+k^nx^n,$$

where $n \in \mathbb{Z}^+$ and $k \in \mathbb{Q}$.

Find the value of n and the value of k. [6]

2. Find the coefficient of x^6 in the expansion of $(2x-5)^9$. [4]

3. Two machines are used in a factory to manufacture semiconductors. Machine A manufactures defective semiconductors 10% of the time, while machine B manufactures defective semiconductors 5% of the time. A randomly selected machine manufactures ten semiconductors. Both machines are equally likely to be selected.

   (a) Find the probability that exactly two semiconductors are defective. [4]

   (b) Given that exactly two semiconductors are defective, find the probability that they were manufactured by machine A. [3]

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4. A company manufactures metal tubes for bicycle frames. The diameters of the tubes, D mm, are normally distributed with mean 32 and standard deviation σ. The interquartile range of the diameters is 0.28.

   Find the value of σ. [5]

5. In a study, measurements for arm span, A cm, and foot length, F cm, are taken from a large group of adults.

For this group, the regression line of F on A is $F = 0.335A - 32.6$, and the regression line of A on F is $A = 2.89F + 99.3$. Each regression line passes through the mean point.

(a) By using an appropriate regression line, find an estimate of the arm span for an adult with a foot length of 19.8 cm. [2]

(b) For this group of adults, find the mean arm span and the mean foot length. [3]

The heights, H cm, of adults in the group can be modelled by a normal distribution with mean 163 cm and standard deviation σ cm. It is found that 88% of the group have a height between 153 cm and 173 cm.

(c) Find the value of σ. [3]